

Excess, Locality and the Meaning Bath: Delimiting Computational Explanations of Consciousness

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Abstract.

Phenomenological consciousness is salient but local, in a very unusual sense, and this can be seen as analogous to a situation that would arise in a many-worlds interpretation of quantum mechanics. To pursue this further, I examine the nature of computation. "Computational formalism" is somewhat of an oxymoron: a material object exceeds any computational formalism, and computation requires an extrinsic process by which symbol strings acquire meanings. Computer science is a hermeneutical discipline. Dead pure syntax interacts with a plenum that is supersaturated with meaning. This allows us to locate a useful complementarity between texts (substrates for interpretation) and matter. New developments in computer science bring these esoteric issues to the fore. Returning to the idea of computational explanation, I discuss "excess" results, paying particular attention to Rice's theorem in recursion theory ("everything is uncomputable"), suggesting how it may be viewed in a discrete "fiber bundle" perspective. These three aspects --locality, excess, and hermeneutics-- come together in the study of mesoscopic physical mechanisms of consciousness.

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1

Locality

1.1 Why is there an 'I'? Call this the $\exists I$ question. Every named first-person attribute that is a candidate for a characteristic of consciousness can be incorrigibly attributed in the third person to other people, and, conversely, a well-made zombie would find use for the first-person pronoun. The $\exists I$ seems to be irretrievably in the domain of the kind of philosophy Heidegger engaged in with the "Why is there anything?" question. He starts by asking us to *feel* the question (Heidegger 1953) and proceed from there.

1.2 If (a) we are required to recontextualize the *feel* of $\exists I$ when we work on this problem, and if (b) we find that causal powers of consciousness are exhausted by its third-person properties, and if (c) we accept that "once you've explained everything that happens, you've explained everything (Dennett 1993:210)," then $\exists I$ studies are best undertaken by social scientists.

1.3 But (b) can be false while (a) is true. Recontextualizing the feel of metaphysical questions seems to be required when discussing the measurement problem of quantum mechanics. It has been asked many times: What does it feel like to be in a superposition? Specifically, take the many-worlds attitude toward the relative state interpretation, assuming some kind of branching topology. This world is path up the universal tree. Which path is *this world*? Why are we in this one?

1.4 We would have to say we are localized along a path. But suppose someone argues we aren't. This could mean two things. This: Maybe we are in a superposition; it is an illusion that the world looks like a sequence of unique outcomes. Or this: The question is meaningless. The latter response does not have to be a rhetorical posture; it is not necessarily giving up. It is not true that we should feel like *this is one contiguous lived-world* if the premises and conclusions of a many-world interpretation were correct. What sort of physical science antecedent would yield the italicized clause above as consequent? But certainly there are philosophical consequences of this clause that can feed back and revise one's interpretation of physical theories.

1.5 Some of these notions may be worth *not* talking about, not out of a dogmatic reductionism, nor out of behaviorism, but simply because there is some aesthetic in minimality. It is not that we take Occam's chainsaw to our cultural conceptions, it is just that we hold them in abeyance. For example, Buddhism provides a fairly worked-out version of a no-soul theory that is isomorphic in feeling to dropping the *this is one contiguous lived world* belief from our system. The text of thoughts seems so tightly woven that experimenting with some loss seems safe; it is not a formal system.

1.6 The demands that many-worlds made of our interpretive abilities is similar to what is demanded in a family of interesting viewpoints on the consciousness problem. Return to 1.1. There seems to be an attribute $I()$ that applies to some objects, say, minds. The attribution $I(m)$ seems to be true for only one mind m in a world with many minds. Attributing I -ness to only one mind is not the same as solipsism. Solipsism claims there is one mind. (This is not a discussion of solipsism; it is rejected.) There is another "claim": *I am localized here*. *I am not in your mind*. How is this done? Could it be undone?

1.7. A toy example. Here is one factor in the universal ket:

$$|\psi\rangle = |\hat{I} \otimes \otimes \dots\rangle + |\otimes \hat{I} \otimes \dots\rangle + |\otimes \otimes \hat{I} \dots\rangle \quad (1)$$

where

$$\hat{I} : 'I' \qquad \otimes : \text{not-}I$$

Each ket enumerates all the minds (mind-Fock space?). As we go through each term in the sum, each mind gets a turn being 'I'. This is not excessively crazy. If one could only employ classical and algorithmic explanations, this is not worthy of scientific talk: "once you've explained everything that happens," etc. But if the universal wave function can be anything at all like this, phenomenological (irreducible, pure-first-person) consciousness might *make things happen*. Because I -localization may have actual physical effects: a subsystem in a state like (1) might show interference. Positing a universe organized as a superposition of monadic I -states could actually be testable. This despite the universality of transfer of predicates from first to third person.

1.8. When you see an expansion of a ket you ask: why *this* basis and not another? Maybe because this basis is the set of eigenstates of an interesting observable. Or maybe it is arbitrary. Occam here: let's say it's arbitrary. Even though the example is a toy, it is not nonsense and this question is important.

1.9. Now make 1.7 a better toy example: the elements labeled minds in the kets are *less than* minds. They are middle-sized pieces of brains, parallel quantum agents, states superimposed and resisting collapse. Orchestrated with full neural mechanisms, we might expect the emergence of minds. Phylogenetically prior, we might see less, but still with flickers of I -ness.

1.10. *Penumbral pandemonium*: There is no conflict if we open the Dennett (1991) multiple drafts models (wrenched --only slightly, despite appearances-- out of its philosophical background) to admit Penrose (1994) *Shadows*-style models. Mesoscopic quantum effects localize bits of I -ness differently and these are superimposed-- that's one factor in the tensor product. Heterophenomenology still supports the bulk of these models. And we could fabricate artificial consciousnesses.

1.11 Quantum mechanics opens the door for lots of things, including a kind of distributed pinealism (a Cartesian pineal gland in every neuron), a dualist theory in which mind-stuff interacts with matter by nudging quantum fluctuations. This may be experimentally indistinguishable from penumbral pandemonium.

1.12. So what does it *feel* like? Are we trapped in our own subminds? Our language itself has chosen a basis. Dangerous territory, because it is all too poetically seductive, and most of us like to remain corrigible. It is natural to follow Dennett and do a Kantian pullback: just as Kant pulled discourse from thing-talk to phenomena-talk, Dennett continues and pull phenomena-talk out to stance-talk. And when each discourse stabilizes, it becomes "realism".

2. Diagonalization and Other Excesses

2.1. Cantor's diagonalization process says just about everything about the limits of formal systems. Rogers (1967): "It is not inaccurate to say that our theory [i.e. recursion theory] is, in large part, a 'theory of diagonalization.'" Self-reference is the source of many limitation results, and it enters through diagonalization: *feed a machine's description to itself*, and so on. "Reason can go beyond whatever reason can formalize (Putnam 1988:117)". There are more things in heaven and earth than are representable in your formalism. This is central to Penrose (1989, 1994).

2.2. We turn to the idea of an algorithm. It is a mathematical notion, and by defining it via a Turing machine, we may pin it down as tightly as, say, the notions of *group* or *vector space*. "*Is our physical space a vector space? No.*" is a dialog about models, and we all know that, and we can be casual about the copula. Literally, the question is meaningless, a category mistake, but we understand the "is" means "can be modeled as," and the context of the question (freshman mechanics text or general relativity text) might determine different answers. The concept of vector space is fixed, and we will not change it as we learn more about physical spaces. This much is familiar.

2.3. But when the notion of algorithm meets the physical world, funny things happen. For example, Dennett (1995) uses a very wide notion of algorithm that encompasses evolution and just about everything imaginable. Is evolution algorithmic? That seems to be a question like "is WW II a stimulus?" in the famous Chomsky refutation of Skinner; the term has widened so much it is useless. "Algorithmic" for Dennett means anything mechanistic with occasional access to randomness. But all he is saying, it seems, is that there is no need to postulate the existence of a real process that could solve halting problems, figure out uncomputable tilings, or do X [where X is some real-world-survival uncomputable problem].

2.4. But when we look at a piece of the world and say "it is executing algorithm A ," we doing interpretation. Is there a canonical interpretation? Conrad and Rosenthal (1981) once suggested measuring the computing power of biological

systems by the complexity of the automata that simulate them, but this view required a great deal of charity. How do we agree on states and dynamics? If you invite five physicists to stand around a table upon which you've placed some kind of mass-and-string contraption, and ask them to write down its Lagrangian, if you get the same Lagrangian from each of them it is only because there are conventions in this situation. So it is certainly valid to measure the computing power of a class of physical systems by the computing power of some formal model that would *conventionally* be used in describing it. (I used to think otherwise, citing Putnam's "Theorem" that every system realizes every finite automaton; see Kirby 1991).

2.5. But the crucial move in the above paragraph is very subtle: "a class of physical systems". Abstraction. It's not the plurality that matters, it's the grouping as a class. Once we've abstracted that way, then we may throw in a convention and the relationship to a mathematical entity (group, vector space, or algorithm). But we could not make the same move with a thing, a lump, a particular individual bit of the individual world, considered by itself.

2.6. Some simpler cases. In mathematical ecology one would write down two simple nonlinear differential equations to model predator-prey dynamics. One then looks at real lions and real gazelles. Sure, they look different; it was only a model. Here is the same situation: in linguistics, some mathematically-oriented linguists like to ask exactly where in the Chomsky hierarchy (regular, context-free, context-sensitive, recursive, recursively enumerable, or uncomputable) is natural language. Some had cited evidence to show that that Dutch was not context-free. Of course, this was an *idealized* Dutch. This same genre of research asks for necessary and sufficient conditions for a language, formally specified, to be a *natural* language, i.e., one learnable by children with realistically small input. Of course, this would be an idealized language and an idealized child. But fixing the idealization *also* required modeling. It is not circular only insofar as the conventions are stable.

2.7. Are our brains executing algorithm *A* when they do *X*? A *class* of brains, an *idealized* brain, a Platonic form of a brain, may be usefully interpreted as having states and state transitions such that our description of them is isomorphic to a Turing machine which, given certain I/O coding conventions, can be seen as doing *X*. Or, if Penrose is right, it may *not* usefully (and usefully *not*) be interpreted as doing so.

2.8. Whether "the brain is doing something uncomputable," as a categorical statement, is true or false, requires there to be some stable, canonical interpretation of brain dynamics. Functionalists like Fodor maintain this is the case (even better for them, they maintain there is a *language* of thought). But Fodor's functionalism is false; Putnam and Penrose agree on that. So the search for the algorithmically uncomputable seems over before it begins.

2.9. Excess is power. The individual always *exceeds* its formalization. This is what makes evolution possible. Evolution depends on *things*. Things evolve, representations do not (Kirby 1995). This bodes ill for simulating evolution. A *mutated description of an evolving system* may indeed result in a *description of a*

mutated evolving system, but true amenability to evolution lies in individual material complexes. Although we have the right to view them as realizations of a system (which allows us to distinguish between "function" and "malfunction", for example), *their power lies in their resistance to our functional descriptions*. The power of the polypeptide folding into an enzyme lies not in the complexity of the energy minimization process, but in the fact that a shape can "go out into the world" and palpate the unknown, playing different roles, whereas a formal string cannot.

2.10. It is fitting that Dennett uses his recent book on evolutionary theory attack (incorrectly, I believe) Penrose's theories of mind. Whether discussing evolution or the brain, a similar issue arises. If evolution is algorithmic, we could simulate it on a computer. Evolution is *not* algorithmic, but that's not because it has access to some mysterious trans-Turing power. (Maybe it does.) It's because any given lump of matter can assume unforeseen functions when interacting with its surroundings. Modeling it fixes its description. If you want to think of evolution as a big computation, then it becomes an empirical issue as to how useful this really is. Very few evolutionary explanations seem computational (*unlike* brain theory), so I think one would believe that evolution is algorithmic only for ideological reasons. (It does harmonize with a certain view of the world.)

2.11. The emphasis on physical things developing new function by interaction with their environment may have laid too much emphasis on the physical. In fact, our linguistic life is made up by metaphors, which, on Rorty's (1991) interpretation of Davidson, don't redescribe something that exists, but instead create something new. Metaphors literalize and enlarge the space of discourse. This is not algorithmic. This fails to be algorithmic *not* because we need to, say, solve the halting problem to come up with new metaphors, but because the interaction with the environment is so vast it is not illuminating for very long to use a computer model for metaphor generation.

2.12. Apropos of the previous paragraph, here is I.A. Richards (1953) on the act of translation (e.g. from English to Chinese):

We have here indeed what may very probably be the most complex type of event yet produced in the evolution of the cosmos.

Indeed. And this is not *computational* complexity!

2.13. The study of the interpretation of matter as a dynamical system doing computation is called *computer science*. Computer science is hermeneutics. At the juncture of things and representations, the material of the world, ambiguous, superimposed, and multi-functional, meets the interpreted logic of formal symbols. Now that we have Turing machines and mesoscopic quantum computers, computer science will become *hard*.

2.14. What does a mature computer science look like? It starts with *texts*, which are substrates for interpretation. Not every text can be interpreted every way (*pace* both Putnam on dynamical systems and Derrida on writing). The dynamics has to

be somehow salient, somehow amenable to a certain interpretation. In addition to the familiar syntactic operations on symbols, we will need to supply new computational primitives: superposition and conformation are two that come to mind. And we also need to discuss complexity, the idea of computing "power." But this computer science must modest: the real world crucially depends on *excess*, on what is just outside the texts.

2.15. Are some texts self-interpreting? I think not. But it is amusing to compare this problem with what we might call the "literary criticism argument for the existence of God" hinted at by George Steiner (1987): God as the uninterpreted interpreter (who existence is tacitly posited by the existence of texts).

2.17. *Digression.* One "modern" mode of computation is superposition. Here is one simple example. Bohm and Hiley (1993), in their thorough critique of the Everett relative state interpretation, zoom in on the problems with representing minds in Hilbert space. In the course of their discussion we read

For in these cases [all but one term in a superposition being zero], the memory would be determined by an eigenvector of what may be called a memory operator.

It is entertaining to note that connectionists have been using memory operators and memories-as-eigenvectors! For example, the "Brain-State-In-A-Box" model (Anderson and Murphy 1986) takes formal neurons $i=1,...,n$ and can store the memories patterns $|\mu_1\rangle, |\mu_2\rangle, ..., |\mu_M\rangle$ by creating the operator:

$$W = \sum_{k=1}^M |\mu_k\rangle \langle \mu_k|$$

or, more realistically, by an iterative process

$$W(0) = 0$$

$$W(t+1) = W(t) + (|\mu_k\rangle - W(t)|\mu_k\rangle) \langle \mu_k|$$

There is a preferred basis $|1\rangle, |2\rangle, ..., |n\rangle$ representing the state where one formal neuron is "on" and the others are "off". The number $\langle i | \mu_k \rangle$ represents the activity level of formal neuron i for memory $|\mu_k\rangle$. The real number $\langle i | W | j \rangle$ represents the strength of the connection from neuron j to neuron i . In the simplest case, if we apply a query input $|\psi\rangle$ and repeatedly feed the state back into the network, obtaining $WW...W|\psi\rangle$, this would converge to the dominant eigenvector of W . By clipping the components to a "box" in the preferred basis, and tinkering with the iteration rule, we can get each of the stored memories back. Anderson has even simulated some simple *modus ponens* inference in a toy "expert system" based on this model. For about ten years I have been using the Dirac notation to teach connectionism; it is useful when you have to give long names to patterns, e.g. $|\text{lock}\rangle |\text{key}\rangle$. It might be interesting some time to take it seriously.

2.18. One last topic on the *excess* issue. One under-appreciated theorem in computer science is a very powerful corollary of the unsolvability the halting problem. This is Rice's Theorem, presented in most computability theory textbooks, though usually without much fanfare (in Rogers 1967 it is relegated to an exercise.) It says, roughly:

Everything in itself is uncomputable.

This is my hyperbolic spin on it, and it is certainly worth clarifying. More precisely, Rice's theorem says *every nontrivial property of the recursively enumerable languages is undecidable*. Let's unpack this.

Fix some alphabet, say $\{0,1\}$. A *formal language* is any set of finite strings built up from this alphabet. We are interested in recursively enumerable languages. A language is *recursively enumerable* (or r.e.) if there is a Turing machine that accepts every string in the language. Denote the set of r.e. languages by $\mathbf{B}$. There are many ways to describe r.e. languages. We could use formal grammars, for example. Here we'll stick to Turing machines, realized in some string representation. If m is [a string representation of] a Turing machine, let $L(m)$ denote the language accepted by the machine [represented by] m . Denote the set of all string representation of Turing machines by $\mathbf{E}$. So $L(m) \in \mathbf{B}$ by definition, and we can even view L as a function $L : \mathbf{E} \rightarrow \mathbf{B}$.

A *property* of the r.e. languages is something like "is finite", or "all it's strings contain the substring 10111", or even "is recursive". These specify subsets of $\mathbf{B}$. A *non-trivial property* is any proper subset $\mathbf{P} \subset \mathbf{B}$. Now Rice's theorem says that *any* non-trivial property $\mathbf{P}$ is undecidable. That is, there is no way to algorithmically decide whether a language has property $\mathbf{P}$ or not!¹

Now how would one even attempt to decide a property of a language. You would need to represent it in some way that an algorithm could digest. Suppose you represented a language λ by a Turing machine. There would be many machines m such that $\lambda = L(m)$. Now it's certainly true that you could decide certain properties of Turing machine strings. If S is an arbitrary set of Turing machine strings this set may very well be recursive. But sets of Turing machines of the form

$$L^{-1}(\mathbf{P}) = \{ m \in \mathbf{E} \mid \lambda = L(m) \text{ for some } \lambda \in \mathbf{P} \}$$

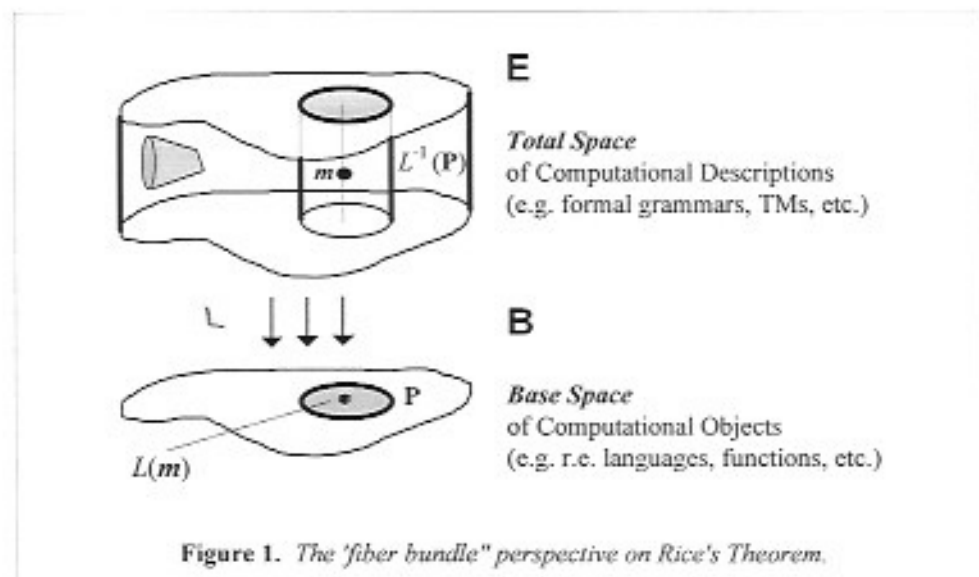
is never recursive. That means $\mathbf{P}$ is never decidable.

So a set of Turing machine descriptions may or may not be computable, but a set of Turing machine that exclusively represent a property of language (including the case of one language by itself) is undecidable. And there is nothing special about languages here; the traditional presentation works this out using functions of natural numbers. Rice's theorem is a diagonalization theorem too; L maps from a set of strings to a set of sets of strings, and there are formal

¹The proof uses a reduction from the halting problem. I may include it in a revision of this MS.

similarities to the familiar proof that the cardinality of the power set is greater than the cardinality of the original set.

2.19. The point of bringing up Rice's theorem was not to salvage an interesting result from exercises in old textbooks, but to bring up a new point of view on computability. Figure 1 illustrates this.



This picture sets up computability as if it had some kind of fiber bundle structure. The base space is the set of all r.e. languages, the total space is the space of all string representations of Turing machines. The projection L takes each Turing machine to the language it accepts. Each fiber $L^{-1}(\lambda)$ is a countable set representing all the Turing machines that accept the same language. Sets that are unions of fibers, like $L^{-1}(P)$, are never recursive (unless equal to E or the empty set).

What if we take this sketch seriously as topology? On the case of discrete syntactic computation, there is no significant structure here. But what if the computational objects in the base space are not languages but dynamics of a certain nature? The base space in this case is indeed a manifold, and sections of the fiber bundle (the fields, as it were) form a preferred computational representation, which are structural descriptions of the system that executes it. A trajectory in the base space may or may not lift to an effectively computable series of transformations of structural descriptions.

3. Coupling with the Meaning Bath

3.1 Despite the I-ness/locality/privatization discussion (section 1), I believe we are still bound by the Kripkensteinian problem (Kripke 1982): there is no fact of the matter that can decide someone is following a rule. There is, consequently, no private language. The solution is to turn to an interpretive community. This is dual to the considerations texts versus individual things in 2.5-2.10.

3.2. A computation requires a substrate, but is quickly exceeded by it. There is a plenum of functionality in the physical world which it can draw on to surpass any computational description. *Obversely*, a computation requires an interpretive environment from which it can receive an interpretation. This environment is a "meaning bath", is a reservoir of meanings. (The analogy with decoherence in some quantum measurement theories is inviting.)

3.3. Meanings are irresistible. Even most recondite sequence of impressions cannot fight off being given meaning.² One might also mention the idea of "naturalization" of texts by readers (Culler 1975), in which a reader evolves a discord-minimizing interpretation of a narrative. The text is coupled to other texts and to the culture. As in a frustrated spin glass, with some poetic texts a unique stable interpretation like this may not happen, and the trajectory of interpretation seems nearly ergodic.

3.4. Another sketch. We have some kind of syntactic generation of a formal object. The formal object is realized (new modes of excess made available), and acquires its meaning through some unforeseen use. The protein is the natural example:

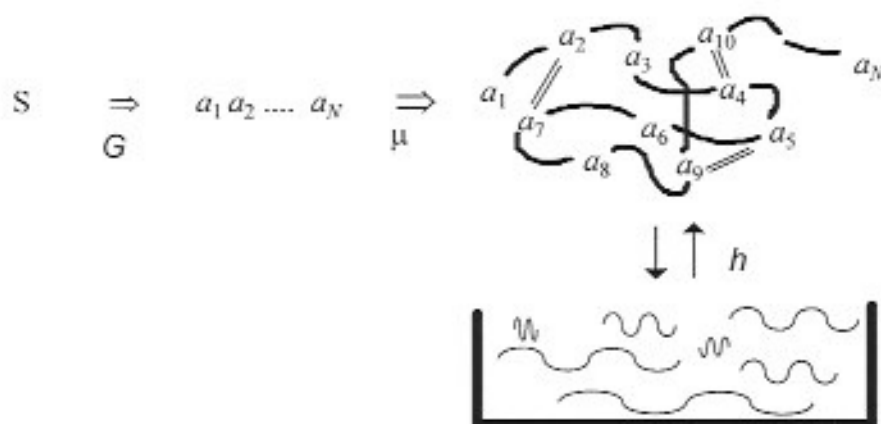


Figure 2. Generation, conformation, interpretation: the metaphor.

²Except maybe these paragraphs.

3.5. Searle 1993:209: "the really deep problem is that syntax is essentially an observer-relative notion."

[...]

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